

Final Examination of Summer Semester 2026
CSC 471 Microprocessor based system (Section: C, D, E)
Course Instructor: Dr Md. Alomgir Hossain
Marks Distribution Rubric

Rubric for Final Assignment — 10 Marks

CLO	Assessment Criteria	Excellent (Full Marks)	Good (Partial Marks)	Satisfactory (Minimal Marks)	Needs Improvement	Marks Awarded
CLO 1: Understand microcontroller interfacing with sensors	Sensor integration and correct data acquisition	2	1.5	1	0	/2
CLO 2: Circuit design and implementation	Circuit diagram correctness and component selection	2	1.5	1	0	/2
CLO 3: Program development and logic flow	Quality of code, logic flow, and commenting	2	1.5	1	0	/2
CLO 4: Simulation/testing and result verification	Simulation correctness or real hardware testing results	2	1.5	1	0	/2
CLO 5: Report writing and explanation of design	Clarity of explanation, flowchart, and report quality	2	1.5	1	0	/2
Total Marks						/10

Notes:

- Total: **10 Marks** — easy to grade.
- Criteria are **aligned with our course outcomes** and my scenario.



Evidence Against Each Criterion

The 'Marks Awarded' column in the rubric above is left blank for the course instructor. This table records where in the report each assessment criterion is demonstrated.

Course Learning Outcome	How it is met in this submission
CLO 1 — Microcontroller interfacing with sensors	MPU6050 interfaced to the ESP32 over I2C at 400 kHz, configured by direct register writes to ± 16 g (2048 LSB/g) and ± 2000 dps (16.4 LSB/dps), with the on-chip low-pass filter enabled as an anti-alias stage and a sample-rate divider of 19 giving exactly 50 Hz. Acquisition runs from a hardware-timer interrupt into a 100×6 ring buffer rather than a polled delay, because loop jitter accumulates and shifts window content away from the training distribution. Report Section V-E and Fig. 1(a).
CLO 2 — Circuit design and implementation	Complete prototype built and validated: ESP32 DevKit V1, MPU6050 on GPIO 21/22, active buzzer on GPIO 4, status LED on GPIO 2, powered from a 1000 mAh LiPo cell through a protected TP4056 charger. Components were selected against an explicit BDT 1,450 budget, and the ± 16 g range was chosen deliberately because fall impacts exceed 8 g and a clipped peak destroys the signal the model depends on. Report Section V-E.
CLO 3 — Program development and logic flow	Two documented, version-controlled codebases. The training library defines preprocessing once and is imported by both the training scripts and the firmware generator, so the two cannot diverge. The firmware implements the timer ISR, ring buffer, unit conversion, per-window instance normalisation, INT8 quantisation, inference, k-of-n agreement and cooldown, and additionally runs a rule-based detector on the same live samples for comparison. Report Fig. 1 and Section IV.
CLO 4 — Simulation/testing and result verification	Verified at three levels. Offline: 5-fold subject-grouped cross-validation, full 25-fold and 32-fold leave-one-subject-out, and leave-one-dataset-out across four corpora (Tables I–IV). Quantisation: FP32 against INT8 on identical held-out subjects, with 99.4 % agreement. On hardware: the firmware instruments its own inference latency and tensor-arena high-water mark. Every split is by subject or by dataset, never by window, to prevent leakage.
CLO 5 — Report writing and explanation of design	Six-page report in IEEE conference format, with a method figure showing both the on-device pipeline and the network architecture (Fig. 1), a results figure quantifying the lead-time versus false-alarm trade-off (Fig. 2), and five tables. Limitations are stated explicitly rather than omitted, including simulated rather than real falls, an incomplete SisFall mirror, and the confounded UMAFall fold.